26900061 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBTG301 - BASIC BIOINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. State one importance of literature mining. [K] / [CO1]
2. What do you mean by a nucleotide sequence database? [K] / [CO2]
3. What type of information is used to build a MODELLER model? [U] / [CO3]
4. Name one type of phylogenetic tree based on branching pattern. [U] / [CO3]
5. What does RMSD measure in Homology Modelling. [U] / [CO3]
6. Identify How does Map Viewer represent genomic information? [U] / [CO3]
7. State the main use of RasMol. [U] / [CO3]
8. Distinguish between SBDD and LBDD. [U] / [CO4]
9. What is confidentiality in bioinformatics? [K] / [CO4]

10. Differentiate between a hydrogen-bond donor and a hydrogen-bond acceptor. [U] / [CO4]

11. What is a binding site? [K] / [CO4]

12. Define the computational study of an organism's complete genetic material. [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain how to retrieve gene information using Ensembl. [A] / [CO2]

14. Explain the major uses of GenBank in bioinformatics. [U] / [CO2]

15. Describe the basic principle of an ORF Viewer. [U] / [CO3]

16. Demonstrate how insertions, deletions and mutations can be detected using dot plots. [S] / [CO3]

17. Explain the role of eDNA and geospatial analysis in biodiversity assessment. [An] / [CO4]

18. Describe the application of bioinformatics in pharmacogenomics and predicting adverse drug reactions. [An] / [CO4]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

19. Describe the Human Genome Project and explain its objectives, methodology and major achievements. [U] / [CO1]

20. Describe the major primary sequence databases and explain their role in storing biological sequence information. [U] / [CO2]

21. Compare the Needleman-Wunsch and Smith-Waterman algorithms with respect to principle, scoring and applications [An] / [CO3]

22. Evaluate the concept of Classical Structure–Activity Relationship (SAR) and describe the major steps involved. [An] / [CO4]

[15x2 = 30]